Kella Srinuprasad

ee22resch11009@iith.ac.in | srinuprasadkella97@gmail.com | +91 9160117541 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
St. Theressa Institute of Engineering and Technology, Garividi, Vizianagaram	Class XII (State Board of Technical Education and Training)	2016	80.71%
Govt. High School Nellimarla, Vizianagaram	Class X (Board of Secondary Education)	2013	8.70%

TECHNICAL SKILLS

PCB designing, Gate driver design, Usage of micro controller launchpads, Circuit implementation using FPGA

SOFTWARES AND PROGRAMMING

MATLAB/Simulink, PLECS, KiCad, Code Composer Studio (CCS), Quartus

PROJECTS

Control of Multi-phase Induction Motor Drives

- Analyzed and simulated the dynamic models of electrical machines
- Implementation of field oriented control

Current Source Inverter (CSI) Topologies

- Designed gate driver for CSIs
- Prepared PCB layout using KiCAD software
- Developed a 10 kVA SiC-MOSFET based CSI

V/f control technique for CSI-fed motor drives

- Implemented a novel current-regulated V/f control scheme
- Performed the experimental verification on 5 kW motor prototype

M. Tech. Thesis: Simultaneous Mitigation of Torque and Power Pulsations in DFIG under Harmonically Distorted Voltage Grid Condtions

B. Tech. Thesis: Direct torque control of BLDC motor using space vector PWM technique

PUBLICATIONS

K. Srinuprasad and J. Titus, "A Hybrid Inverter Drive System for a Six-Phase Induction Motor," Indian Patent Application No: 202641089313, **filed on 22-07-2026**

K. Srinuprasad and J. Titus, "An H7 Current Source Inverter based Current Regulated V/f Controlled Induction Motor Drive,", submitted manuscript to the IEEE Transactions on Industry Applications (**under review**).

K. Srinuprasad, J. Titus and K. Sivakumar, "An Asymmetrical Dual-Three-Phase Induction Motor Drive using Current and Voltage Source Inverters for Traction Applications,", submitted manuscript to the IEEE Transactions on Transportation Electrification (**under review**).

K. Srinuprasad, et al., "A Modified Cascaded Dual Three-Phase Current Source Inverter for Common-Mode Voltage Reduction," *2026 IEEE International Conference on Power Electronics, Drives and Energy Systems (PEDES)* , Guwahati, India, 2026 (**To be presented**).

K. Srinuprasad, et al., "Comparative Analysis of Commutation Protection Time Effects on Efficiency and Harmonic Distortion in SiC-based Current Source Inverters," *2026 Energy Conversion Congress & Expo Europe (ECCE Europe)*, Valencia, Spain, 2026 (**To be uploaded in IEEE xplore**).

K. Srinuprasad and J. Titus, "Current-Regulated V/f Control for an Induction Motor using SiC-based Current Source Inverter," *2025 IEEE Industry Applications Society Annual Meeting (IAS)*, Taipei, Taiwan, 2025, pp. 1-6, doi: 10.1109/IAS62731.2025.11061526.

K. Srinuprasad and J. Titus, "Sensorless Field Oriented Control for an Induction Motor Drive Using an Ideal Voltage Integration Scheme with a Dynamic Stabilising Feedback," *2023 IEEE Vehicle Power and Propulsion Conference (VPPC)*, Milan, Italy, 2023, pp. 01-06, doi: 10.1109/VPPC60535.2023.10403146.

ACADEMIC ACHIEVEMENTS

Awarded "**Travel grant** to attend ECCE Europe 2026 Conference" by IEEE power electronics society

ACADEMIC RESPONSIBILITIES

Worked as **Teaching Assistant** for electrical machines theory and lab courses

Mentored graduate and post-graduate students in power converter testing, hardware debugging and designing of PCB-prototypes

RELEVANT COURSES

- Power Converter Design
- Modeling and Control of Electric Drives
- Embedded System Design in Power Electronics

EXTRACURRICULAR ACTIVITIES

Participated in competitive sports, representing the **NIT Silchar Cricket Team** in an fixture against SBI Zonal Team